Benchmarking Counterfactual Explanations of Model Change with Exact Ground Truth

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Abstract. Methods that explain a model update are rarely checked against ground truth. We build a benchmark with exactly known changes, null updates included, and a linear analysis of what five families of explanations detect, among them differential counterfactuals (Δ-CE): the nearest inputs on which only one model changes its decision, both decisively. As predicted, gradient differences miss threshold shifts and detect rotations on small CNNs, though not on a ResNet-8. Δ-CE stays silent on gently fine-tuned null updates, where counterfactual and attribution differences report change, and localises two independent changes; a pre-specified hypothesis that it names the changed factor more often than gradient differences was rejected. Its silence comes from the margin check; a post hoc analysis shows that the search adds witnesses a filtered ordinary counterfactual misses.

1 Introduction

An update that improves accuracy can still change individual decisions in unwanted ways [1]. Explaining *how* the models differ is a goal of explainable AI under distributional change [2, 3], distinct from explaining a shift of the data [4]. Artelt et al. [5, 6] compare the counterfactuals [7] of one input under each model, or take a difference of gradients; Muschalik et al. [8] compare feature importances; disagreement search [9, 10, 11] looks for inputs on which the models disagree. Ground-truth benchmarks exist for explaining one model [12, 13, 14], not a change, so what each family can see goes unchecked.

We say that a method *sees* a change if it stays silent when nothing changed (abstention), reports a change when there is one (*whether*), and names the changed factor and region (*what*, *where*). As a fifth family we study *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision, both models by a margin κ. Unlike robust counterfactuals [15, 16], valid for both models, it is valid for exactly one.

Contributions. (i) A benchmark with exact ground truth: five rendered factors, four shift types, null updates and a 4×4 grid over rotation and threshold. (ii) A linear analysis separating rotation from offset, tested on CNNs and a ResNet-8 under a protocol fixed in advance; all failures are reported. (iii) Δ-CE stays silent on gently fine-tuned null updates, also on unscreened nulls and through a learned decoder, where counterfactual and attribution differences report change; it localises two independent changes, and finds witnesses that a filtered counterfactual of f_{new} misses.

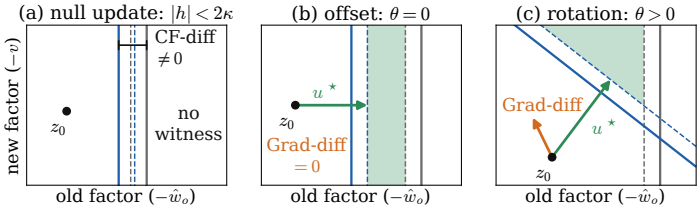


Fig. 1: Linear case. Solid: boundaries of f_{old} (grey), f_{new} (blue); dashed: $m_o = \kappa$, $m_n = -\kappa$; shaded: witnesses. (a) None exists (Lemma 1). (b), (c) Gradient difference and witness $u^\star$ (Proposition 2).

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For a query z_0 that both models assign to class y, the margin is the logit difference $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$; z_0 is valid if $m_o(z_0), m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|_2$ if $\|z^\star - z_0\|_2 \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models disagree, both decisively.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$, so models whose margins differ by less than 2κ everywhere have none.

Corollary 1 (constant logit offset). If f_{new} adds b to the logit of class 1, e.g. the prior correction [17], then $h \equiv \pm b$ and $\nabla h \equiv 0$. Type-N witnesses exist only for the class whose margin b lowers and only if $|b| \geq 2\kappa$, at the points with $\kappa \leq m_o \leq |b| - \kappa$. *Proof:* insert $h \equiv -|b|$ in Lemma 1; no linearity is needed.

Linear case (Fig. 1)**.** For linear margins write the unit normals as $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$, $v \perp \hat{w}_o$ ($\theta = 0$: offset; $\theta > 0$: rotation towards v). Let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$, u_n, u_o the steps to the minimum-norm counterfactuals of f_{new}, f_{old}. We compare gradient difference [5] $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing $u_n - u_o$ and witness $u^\star = z^\star - z_0$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has a larger component along v than along $\hat{w}_o$ for every θ, and so does the witness iff $\theta > \pi/4$ or $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$, all three coincide up to positive scale, as in Theorem 1 of [5] (new here: the offset case and the witness). *Proof* of (a): both normals equal $\hat{w}_o$; of (b): online supplement.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0,1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise. The exact renderer gives

Table 1: Adaptation conditions; $k^\star$: ground-truth factor.

Change	Shift	Adaptation data	$k^\star$	Pairs
Shortcut	spurious	colour correlated with label (0.99)	colour	3
Subgroup	local concept	labels flipped for size > 0.75	size	3
Threshold	concept	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	none / covariate	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid[a]	concept	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$	aspect / colour	8 / 24
Offsets[b]	concept	grid, $\alpha = 0$, $t \in \{.53, .56, .6\}$, seeds 2–4	aspect	9
Both[c]	spurious + local	Shortcut and Subgroup at once	colour, size	8

[a] $\alpha \in \{0, .1, .25, .5\} \times t \in \{.5, .53, .56, .6\}$, 2 seeds; 4 offsets tested. [b] 8 qualified. [b,c] Registered.

the factor change of any explanation without a learned oracle, and all methods search the same factor space through it. f_{old} is a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1); a higher label threshold is an offset and a rule that adds colour a rotation. A pair enters only if its change is visible in decisions on validation data. The ground truth is the injected change; what f_{new} learns is read from the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of a logistic fit to its decisions, at which the grid is analysed.

Metrics and methods. On $N = 300$ valid test points per pair, Recovery@r is the share with a valid explanation within the budget r whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0); the found-rate is the share with any valid explanation within r. For Δ-CE we count type-N witnesses only (pre-declared), so a query whose class the update raises counts 0. All counterfactual methods share one solver, the test points and the budget $r = 0.54$, set once on validation. A margin score, $|h(z_0)|$, serves as the control for *whether*. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [5]; **Grad-diff**, the gradient difference [5]; **Imp-diff**, a difference of permutation importances [8], one vector per pair; **Seg-search**, a line search towards training points, after [9]; **No-prox**, Δ-CE without proximity, the DeepXplore objective [10], to which DiG-IN [11] corresponds on an exact manifold; **CF-new**, the counterfactual of f_{new}; and, in Section 4.2, IG- [18] and SHAP-differences [19].

Protocol. Each test set served at most one confirmatory test, after all choices were fixed. The main test set carries Table 2; hypotheses formed after it were tested on new data, and three follow-ups were registered before their data existed [20]: unscreened nulls, offsets and a ResNet-8; the learned decoder; and two changes at once. Bootstrap intervals resample points within pairs, so Table 3 also counts pairs; the supplement repeats each test with the pair as unit and dates all test sets. A digit-image task and a label-shift update failed qualification.

4 Results

4.1 Abstention, detection and localisation

Abstention. No null query has a witness: on the 8 null pairs of Table 2, five null updates of an MLP on Adult, and 8 registered unscreened null pairs, admitted without the qualification check. On the same nulls (covariate-shifted included) CF-diff reports a change for 97.7–100% of points. Silence does not need the exact

Table 2: Recovery on the main test set (↑) and false reports on null updates (↓).

Condition	Δ-CE	CF-diff	Grad-diff	Imp-diff	Seg-search	No-prox	CF-new
Shortcut ↑	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup ↑	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold ↑	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**[a]	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null ↓	**0.000**	0.994	1.0	1.0	**0.000**	**0.000**	0.996

Change rows: Recovery@r (found-rate), mean of 3 pairs, CIs ±0.035. Null: found-rate on 8 null pairs; any output is a false report. Bold: best in row. [a]Ranks aspect first in every pair.

Table 3: Tests and their outcomes (✓ confirmed, × not confirmed).

Claim	Result	Data	Estimate	p	Pairs
Abstention, detection and localisation					
No witness on nulls[d]	✓ post	Nulls; Adult MLP	0/2 400; 0/1 500	–	8/8; 5/5
Same, unscreened; via decoder	✓ OSF	Registered nulls	0/2 400; 0/800	–	8/8; 8/8
No witness on ResNet-8 nulls	× OSF; expl.	$\kappa = 0.5$[¶]; $\kappa^\star = 2.5$	180/200; 0/200	–	0/2; 2/2
CF-diff flags $\lvert b\rvert < 2\kappa$ [d]	✓ post	Logit offsets, uncut	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$ §	–
Witness ≥ margin score +0.2	× OSF	Registered offsets	+0.17 to +0.97	–	7/8
Witness localises ($L > 0$)	✓ OSF	Two changes	$L = 0.50$	0.004[s]	8/8
Offsets					
Grad-diff Recovery < 0.2	✓[‖] pre	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$ ‡	4/4
Grad-diff Recovery < 0.2	✓ OSF	Registered offsets	0.033–0.093	0.004[s]	8/8
Grad-diff offset-invariant	× pre	Grid	−0.098 [−0.143, −0.053]†	0.94 ‡	–
Δ-CE > Grad-diff [d]	✓[‖] post	Threshold	+0.31 [0.27, 0.35]	$2{\cdot}10^{-4}$ *	3/3
Rotations					
Grad-diff rises with α	✓ pre	Grid	+0.57 [0.53, 0.61]	$4{\cdot}10^{-4}$ ‡	$\rho = 0.89$
Grad-diff < 0.2; rot. > offset	× OSF	ResNet-8	0.32–0.36; 0.06–0.10	–	0/3; 0/3
Proj. Δ-CE ≈ Grad-diff	× post; pre	Shortcut; grid	−0.30; −0.35 (±0.03)†	1 *; 1 ‡	–
Normal at $z^\star$ halves the gap	✓[‖] post	Grid	−0.19 → +0.03 [0.01, 0.05]†	$6{\cdot}10^{-4}$ §	4/4
Δ-*CE versus gradient differences*					
Δ-CE > Grad-diff (rejected)	× pre	Subgroup	−0.09 [−0.13, −0.05]	–	0/3
Δ-CE ≈ Grad-diff, CF-diff	× pre	Shortc., Thresh.	$\lvert\Delta\rvert$ = 0.17–0.53	–	–

Result: outcome and when the test was fixed: OSF = registered [20] before its data existed; pre = before any result, or before the grid run; post = after earlier results, then tested on new data; expl. = exploratory. Estimate: null queries with a witness (count), a Recovery or found-rate, or a difference Δ of Recoveries (≈, invariant: $\lvert\Delta\rvert \le 0.1$); 95% CIs (†90%). p: Holm-adjusted within families *, ‡, §; [s]exact sign test. Pairs: model pairs on the claimed side / all (ρ: rank correlation); [‖]confirmed at point level only (≤ 4 pairs, pair-level $p \ge 0.0625$). [d]Follows from a lemma or the definition. [¶]Run on a local CPU after its cost rule excluded it. Proj.: $\hat{w}_o$ projected out.

renderer: searching through the decoder of an autoencoder trained on the images (registered), Δ-CE finds no witness for a null query (0/800; CF-diff 85–94%), and on 6 of 8 offsets finds one for at least half of the lowered-class queries.

With $\kappa = 0$, witnesses appear on 83% of null queries. Silence also depends on the recipe: the nulls are fine-tuned more gently than the offsets (fewer epochs, lower learning rate; supplement), though their threshold moves as far ($\lvert\Delta\hat{t}\rvert \le 0.035$); the grid's null cell, fine-tuned like the offsets, has witnesses on 22% of queries at $\kappa = 0.5$ and none at $\kappa = 1$, where small offsets keep few witnesses (0.04). Silence is thus a property of κ relative to the size of the update, as Lemma 1 states, and not of the search.

Detection. With each score cut on the nulls to flag none, witnesses exist for 0.17–1.00 of lowered-class queries on the 8 registered offsets, the margin score flags 0.00–0.54 and CF-diff none. The witness beat the margin score by 0.2 in 7 of 8 pairs, short of the registered 8/8, so we claim no advantage for *whether*.

Localisation. With a colour shortcut everywhere and a label flip for large

objects, witnesses change size most for 46–75% of queries near the large-object region and 13–23% far from it; this difference L is positive in all 8 pairs (pooled 0.50; Table 3). Grad-diff and CF-diff barely localise ($L = 0.09$ and 0.13), and Imp-diff, one vector per pair, cannot; CF-new localises equally well (0.51).

Search versus filtering (post hoc, saved outputs). The counterfactual of f_{new} kept only where $m_o \geq \kappa$ (filtered CF-new) is a witness; it is silent on the nulls and localises ($L = 0.48$), but finds witnesses for only 0.42–0.46 of the main-test queries, against 0.47–0.84 for Δ-CE. The extra witnesses carry Δ-CE's Recovery on Shortcut (0.38 vs. 0.03) and part on Subgroup (0.53 vs. 0.30), higher in 3 of 3 pairs each, but not on Threshold (0.47 vs. 0.45). Null-adjusted (minus the rate of naming $k^\star$ on nulls), Δ-CE leads on Subgroup and Threshold, Grad-diff on Shortcut (0.79); unfiltered CF-new falls to -0.09 on Threshold. The search costs 2.4 s per query (CF-new: 0.6 s; Grad-diff: under 1 ms).

4.2 Offsets, rotations and further baselines

Gradient differences are blind to an offset, as Proposition 2(a) predicts: their Recovery is 0.081 on the grid's pure offsets and 0.03–0.09 on the 8 registered offsets (Table 3). On Threshold, Δ-CE beats them on new data (0.41 vs. 0.10; main test set: 0.47 vs. 0.07) in 3 of 3 pairs, too few for a pair-level test. Type-N witnesses exist only for the class the offset lowers (class 1), hence a found-rate of 0.47; type-O witnesses for the other class raise Recovery to 1.00 on new data.

Gradient differences see a rotation, increasingly with its angle: Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ ($t = 0.5$; Spearman $\rho = 0.89$ over the 24 rotation pairs; map of all pairs in the supplement). The projection equivalence of Proposition 2(b) holds only with $\hat{w}_o$ taken at the witness (Table 3).

Attribution differences and other baselines. With predictions fixed beforehand, IG- and SHAP-differences recover colour on Shortcut (0.93) and, against the predicted blindness, name aspect on Threshold (0.73–0.75 vs. 0.07), but also on 57% and 71% of null queries (IG-diff on 42% of the registered nulls, as predicted): no signal. Imp-diff ranks aspect first in all 9 change pairs on test data, $k^\star$ on adaptation data. Seg-search with 10× the segments finds witnesses for 0.26–0.47 of queries (predicted ≤ 0.2), none that Δ-CE misses.

Second architecture. On a registered ResNet-8, null fine-tuning moves margins four times more, and witnesses appear on 0.90 of null queries: κ is model-specific. With κ calibrated on validation nulls (rule fixed in advance; grid extended after $\kappa \leq 2$ failed), $\kappa^\star = 2.5$, no new null query has a witness (0/200) and witnesses exist for 0.78–1.00 of lowered-class offset queries (Recovery 0.36–0.40). Grad-diff sees neither change: on 2 null, 3 offset and 4 rotation pairs it names aspect on 0.32–0.36 of offset queries, near 0.20–0.30 on nulls, and its rotation Recovery, 0.06–0.10, is below the matched offset in all 3 pairs.

5 Discussion and limitations

Our results suggest that an explanation of change needs null updates as a reference, since counterfactual and attribution differences report change on them;

on small CNNs, gradient differences also miss offsets. Abstention comes from the check $m_o \geq \kappa$, which can also filter the counterfactual of f_{new}; specific to Δ-CE is the search for witnesses at queries where that counterfactual fails the check. **Limitations.** (1) The data are synthetic and binary; Recovery through the learned decoder is approximate and lower for Δ-CE than for CF-diff; multi-class tasks, other domains, retraining from scratch and architecture changes are untested, and we did not test why the linear prediction fails on the ResNet-8; the tabular test covers abstention only. (2) κ is in logit units, exact only for a constant offset, and must be calibrated on null updates of each model family and fine-tuning recipe, obtained by fine-tuning f_{old} on resampled data. (3) Witnesses see only decisive changes on the manifold within the budget, without a certificate. (4) Several tests were formed after earlier results; changes were logged before their data existed, with three disclosed exceptions. Online supplement and code: github.com/Johnlrr/explaining-model-change.

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